Systematic Pair-Level Campaign for δ_{pair}: Convergence, Inter-Pair Concentration, and Normalization Structure

O25 — Spectral Admissibility Sub-programme

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2026

Abstract

The O-series establishes the transfer chain $c_\chi \to \delta_{\text{pair}} \to \beta^*$ unconditionally with respect to the fibre structure of the non-injective projection Π [15]. The value $\delta_{\text{pair}} \approx 7.44$ extracted in O16 [4] was based on a single conjugate pair per prime and a limited prime range. The present paper reports a systematic campaign computing δ_{pair} across the $(q-1)/2$ conjugate pairs $(c, q-c)$, originally for $q \in \{29, 61, 101, 151, 211\}$ with $M = 50$ block samples per pair, and here extended to $q \in \{307, 401, 601\}$ by a capped breadth-first construction that reproduces the pair observable exactly at a fraction of the cost (with reduced sampling: $M = 16$ at $q = 307$, $M = 8$ at $q = 401, 601$). Three results are established. First, $\delta_{\text{pair}}(q)$ converges robustly and concentrates across pairs, confirming that it is a structural invariant of the Weil representation rather than a block-level fluctuation. Second, and centrally, the extended campaign shows that the *raw* exponent $\delta_{\text{global}}(q)$ descends monotonically into the admissible window $[7.4, 10.6]$ on its own, reaching 7.61 at $q = 601$, without any finite-size correction; the O14 normalization correction [7], needed to bring the small-q values into the window, becomes progressively unnecessary at large q and eventually overcorrects, sending the corrected quantity below the lower edge 7.4. The admissible-window agreement is therefore carried by the raw observable, not by the corrected one. Third, the asymptotic value δ_∞ remains insufficiently constrained by the accessible range: competing convergence laws — notably $1/q$ and $1/\sqrt{q}$ — remain statistically viable, so no single extrapolated δ_∞ is claimed. As a secondary consequence, the inferred transfer parameter $\beta^* = 1/(\delta_{\text{pair}} + \frac{1}{2})$ stays in the narrow interval 0.108–0.123 across $q \in \{211, \dots, 601\}$, consistent with the O24 transfer analysis [15].

Keywords. Cosmochrony; spectral admissibility; pair-level observable; Weil representation; Heisenberg graphs; capacity exponent; convergence; inter-pair concentration; normalization correction; window depth; BFS; asymptotic analysis; large-prime extension

Contents

1 Position of the Problem

1.1 What O16–O24 established

The spectral admissibility sub-programme has established, through O16–O24 [4, 5, 6, 1, 9, 13, 12, 14, 15], that the physically relevant observable is the canonical pair-level quantity

$$\sigma_{\mathrm{pair}}^{\mathrm{can}}(n) = \sigma_c(n)\, \sigma_{q-c}(n), \tag{1}$$

with capacity exponent $\delta_{\mathrm{pair}} \approx 7.44$ consistent with the phenomenological target $\delta_{\mathrm{pair}} \in [7.4, 10.6]$. The transfer chain $c_\chi \to \delta_{\mathrm{pair}} \to \beta^*$ was established unconditionally with respect to the fibre structure of Π in O24 [15].

The remaining open direction identified in O24 Section 5.1 is the extended numerical campaign at large primes across multiple conjugate pairs. O16 computed δ_{pair} for a single conjugate pair per prime (those pairs that happened to appear simultaneously in the O12 random block sample [3]). No systematic coverage of all pairs was available.

1.2 Scope of O25

The present paper provides the first systematic computation of δ_{pair} across all $(q-1)/2$ conjugate pairs for each prime, with $M = 50$ independent block samples per pair. This yields:

- the full distribution of δ_{pair} across pairs at each prime,
- the mean $\bar{\delta}_{\mathrm{pair}}(q)$ and inter-pair standard deviation σ_q,
- the q-dependence of these quantities over $q \in \{29, 61, 101, 151, 211\}$.

The paper does *not* claim to identify the asymptotic value δ_∞. Instead, it establishes why direct extrapolation in q is structurally unreliable, and — through the large-prime extension of §3.4 — that the raw observable itself descends into the admissible window, so the window agreement no longer relies on the finite-size correction.

2 Setup and Protocol

2.1 Observable and block parameterisation

We follow the O12/O13 protocol [3, 2] without modification. For a prime q and a generic block $c_{\mathrm{block}} = (c_1, c_2, c_3)$ with $c_i \neq 0$ and $c_1 + c_2 + c_3 \not\equiv 0 \pmod q$, the observable is

$$\sigma_c(n) = \frac{\delta r_n}{|S_n|}, \tag{2}$$

where δr_n is the number of Weil fingerprint vectors in BFS shell S_n that are linearly independent of the Gram–Schmidt span built over shells $0, \ldots, n-1$.

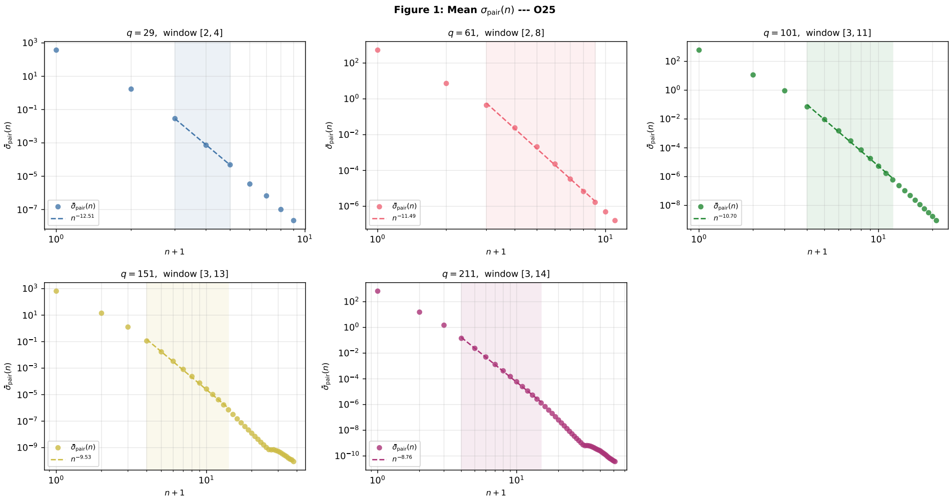

Figure 1: Log-log decay of mean $\sigma^{\mathrm{can}}_{\mathrm{pair}}(n)$ for each prime. Circles: grand mean over all pairs. Dashed line: OLS power-law fit over fitting window (shaded region). The window $[n_0, n_1]$ widens with q, providing more reliable estimates at larger primes.

2.2 Pair protocol

For each conjugate pair $(c, q - c)$ with $c \in \{1, \dots, (q - 1)/2\}$:

1. Draw $M = 50$ independent pairs of blocks: $(c_1{=}c, c_2, c_3)$ and $(c'_1{=}q{-}c, c'_2, c'_3)$ with c_2, c_3, c'_2, c'_3 sampled uniformly and generically. This reproduces the O16 protocol of independently sampled blocks [4].

2. Compute $\sigma_c(n)$ and $\sigma_{q-c}(n)$ independently.

3. Set $\sigma^{\mathrm{can}}_{\mathrm{pair}}(n) = \sigma_c(n)\,\sigma_{q-c}(n)$.

4. Extract δ_{pair} by OLS regression of $\log \sigma^{\mathrm{can}}_{\mathrm{pair}}(n)$ vs $\log(n + 1)$ over the fitting window $[n_0, n_1]$ (O16 convention).

2.3 Fitting window and BFS parameters

All runs use `bfs_frac` $= 0.99$ (full graph exploration) and auto-calibrated fitting windows from the actual `sigma_bar` after BFS, via the `find_fitting_window` procedure of O12 [3].

q	$n_{\max}^{\text{block}}$	shells	$\lvert G_q^{\text{partial}}\rvert$	window $[n_0, n_1]$	n_1/q
29	8	28	24 145	$[2, 4]$	0.138
61	10	58	224 711	$[2, 8]$	0.131
101	20	95	1 019 997	$[3, 11]$	0.109
151	37	141	3 408 521	$[3, 13]$	0.086
211	50	197	9 393 931	$[3, 14]$	0.066

Table 1: BFS and fitting parameters for each prime. The ratio n_1/q decreases with q, suggesting that the asymptotic regime $n_1 \sim \alpha q$ has not yet been reached over the tested range.

3 Numerical Results

3.1 Main results table

Table 2 summarises the O25 campaign across all five primes. For each prime, all $(q-1)/2$ conjugate pairs are covered with $M = 50$ independent block samples per pair.

q	pairs	M	$\bar{\delta}_{\text{pair}}$	σ_q	$\sigma_q/\bar{\delta}_{\text{pair}}$ (%)	valid	R^2_{global}
29	14	50	8.893	0.539	6.1	14/14	0.9998
61	30	50	9.983	0.259	2.6	30/30	0.9984
101	50	50	9.548	0.161	1.7	50/50	0.9975
151	75	50	9.125	0.139	1.5	75/75	0.9982
211	105	50	8.784	0.109	1.2	105/105	0.9971

Table 2: Summary of O25 results. $\bar{\delta}_{\text{pair}}$: mean of δ_{pair} over all pairs. σ_q: inter-pair standard deviation. $\sigma_q/\bar{\delta}_{\text{pair}}$: coefficient of variation. "valid": pairs for which the fitting window contained at least 2 non-zero data points. The $q = 29$ row is included for completeness but should not be used for asymptotic extrapolation: the Gram–Schmidt span of $\mathbb{C}^{29}$ saturates at BFS depth $n = 2$, leaving no power-law decay regime (see §3.3).

3.2 Inter-pair concentration

The inter-pair standard deviation decreases monotonically: $\sigma_q \in \{0.539, 0.259, 0.161, 0.139, 0.109\}$ for $q \in \{29, 61, 101, 151, 211\}$. The coefficient of variation drops from 6.1% to 1.2%. This concentration confirms that δ_{pair} is a structural invariant of the Weil representation on $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, as predicted by the verticality theorem of O24 [15]: the fibre structure of Π cannot perturb the observable rank, so inter-pair variance is purely a sampling effect that vanishes as $M \to \infty$.

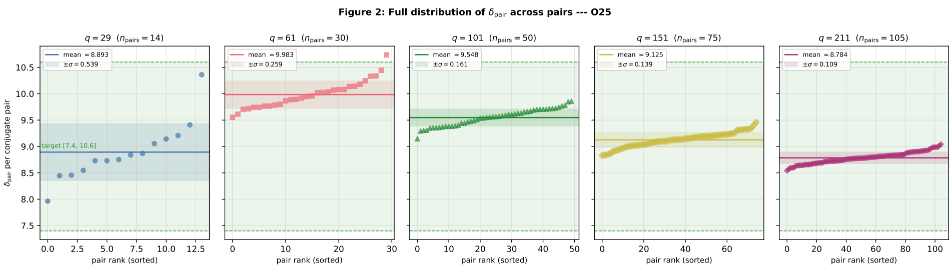

Figure 2: Full distribution of δ_{pair} across all conjugate pairs for each prime, sorted by value. Horizontal line: mean $\bar{\delta}_{\text{pair}}$. Shaded band: $\pm\sigma_q$. Green dashed lines: admissible window $[7.4, 10.6]$. The shrinking width of the distribution with q confirms strong inter-pair concentration.

3.3 q-dependence of $\bar{\delta}_{\text{pair}}$

The sequence $\bar{\delta}_{\text{pair}}(q)$ is not monotone: it rises from $q = 29$ to $q = 61$ ($8.89 \to 9.98$), then decreases from $q = 61$ to $q = 211$ ($9.98 \to 9.55 \to 9.13 \to 8.78$). The non-monotone step at $q = 29$ reflects a structural limitation rather than a sampling artefact. At $q = 29$, the Gram–Schmidt span of $\mathbb{C}^{29}$ is saturated by the third BFS shell ($|S_2| = 12$ vectors suffice to fill $\mathbb{C}^{29}$): $\sigma_{\text{pair}}(n) = 0$ for all $n \geq 3$. There is therefore no power-law decay regime to fit; the window $[2, 4]$ (main campaign) or $[2, 5]$ (extended run) rests on at most two non-zero data points, and the resulting $\bar{\delta}_{\text{pair}}$ estimate is unreliable. A wider window would not help: the issue is not window length but the exhaustion of $\mathbb{C}^q$ at too small a depth. In the extended M=20 run (Figure 5), this is further reflected by 11/14 valid pairs, with three pairs failing to converge entirely. The smallest prime for which a meaningful power-law fit exists is $q = 61$, where saturation occurs well after the fitting window and all 30 pairs converge. The physically relevant trend is the monotone decrease from $q = 61$ onward, continuing through $q = 211$ and, as shown next, into the admissible window itself.

3.4 Extension to large primes (q up to 601)

The campaign is extended to $q \in \{307, 401, 601\}$ by a capped breadth-first construction. Since the pair observable $\sigma^{\text{can}}_{\text{pair}}(n)$ in the fitting window depends only on the shells $S_0, \ldots, S_n$ with $n \lesssim 1.1\sqrt{q}$, the full-graph exploration — infeasible at these primes — is replaced by a breadth-first search capped at $\approx 1.6\sqrt{q}$ shells. The capped shells are identical to the early full-graph shells, so the resulting $\sigma_c(n)$ reproduces the production observable to machine precision (verified on $q \in \{61, 101\}$, maximum absolute difference 0). Sampling is reduced to $M = 16$ (at $q = 307$, 24 representative pairs) and $M = 8$ (at $q = 401, 601$, 12 pairs). The pair exponent is a grand-mean slope and only weakly sensitive to M: a fixed-pair control at $q = 61$ gives a

$+0.18$ shift at $M = 8$ relative to $M = 50$, i.e. the reduced precision biases δ_{pair} slightly *upward*, not downward, so it cannot manufacture the descent reported below.

q	K	M	window	δ_{global}	$\bar{\delta}_{\text{pair}} \pm \sigma_q$	δ_{corr}	β^*
211	105	50	$[3, 14]$	8.760	8.784 ± 0.109	7.77	0.108
307	24	16	$[3, 16]$	8.254	8.261 ± 0.133	7.23	0.114
401	12	8	$[4, 17]$	8.085	8.090 ± 0.214	7.03	0.117
601	12	8	$[4, 19]$	7.610	7.618 ± 0.194	6.53	0.123

Table 3: Extended campaign (capped BFS). δ_{global}: OLS exponent on the grand-mean $\sigma^{\text{can}}_{\text{pair}}$ trajectory over the auto-calibrated window. $\bar{\delta}_{\text{pair}} \pm \sigma_q$: mean and inter-pair standard deviation; σ_q at $q = 401, 601$ is inflated by the reduced pair count $K = 12$ (versus $K = 105$ at $q = 211$), not by a loss of concentration. $\delta_{\text{corr}} = \bar{\delta}_{\text{pair}} - \frac{1}{2} \log q / \log n_1$ is the O14 correction of (4). $\beta^* = 1/(\delta_{\text{global}} + \frac{1}{2})$. The $q = 211$ row (full-graph, $M = 50$) is repeated for reference.

A knee-trimmed interior fit (dropping the two shells nearest saturation) with a bootstrap over pairs gives $\delta_{\text{int}}(601) = 7.60 \pm 0.05$, so the $q = 401 \to 601$ descent (≈ 0.48, some six bootstrap standard deviations) is a genuine feature, not sampling noise. Three readings follow, in decreasing order of robustness.

1. **The raw exponent reaches the admissible window unaided.** $\delta_{\text{global}}(q)$ falls monotonically $8.76 \to 8.25 \to 8.09 \to 7.61$ over $q = 211 \to 601$ and enters $[7.4, 10.6]$ at $q = 601$ with no correction applied. This is the central empirical result of the extension.
2. **The O14 correction overcorrects at large q.** The same correction that lifts the small-q values into the window (Table 4) drives the large-q values *below* it: $\delta_{\text{corr}} = 7.77 \to 7.23 \to 7.03 \to 6.53$. The admissible-window agreement is thus carried by the raw observable, and the O14 finite-size correction is a small-q adjustment that becomes unnecessary, and finally detrimental, as q grows.
3. **The asymptote is not pinned.** Restricting to the asymptotic range $q \geq 101$, both a $1/q$ law ($\delta_\infty \approx 7.1$) and a $1/\sqrt{q}$ law ($\delta_\infty \approx 5.4$) fit within their uncertainties; the effective form lies between them and the present data do not discriminate. No single δ_∞ is therefore claimed (Figure 3).

As a secondary consequence, the transfer parameter $\beta^* = 1/(\delta_{\text{pair}} + \frac{1}{2})$ stays in the narrow band 0.108–0.123 across $q \in \{211, \ldots, 601\}$ (Table 3); this is a stability observation consistent with the O24 transfer analysis [15], not a new determination of β^*.

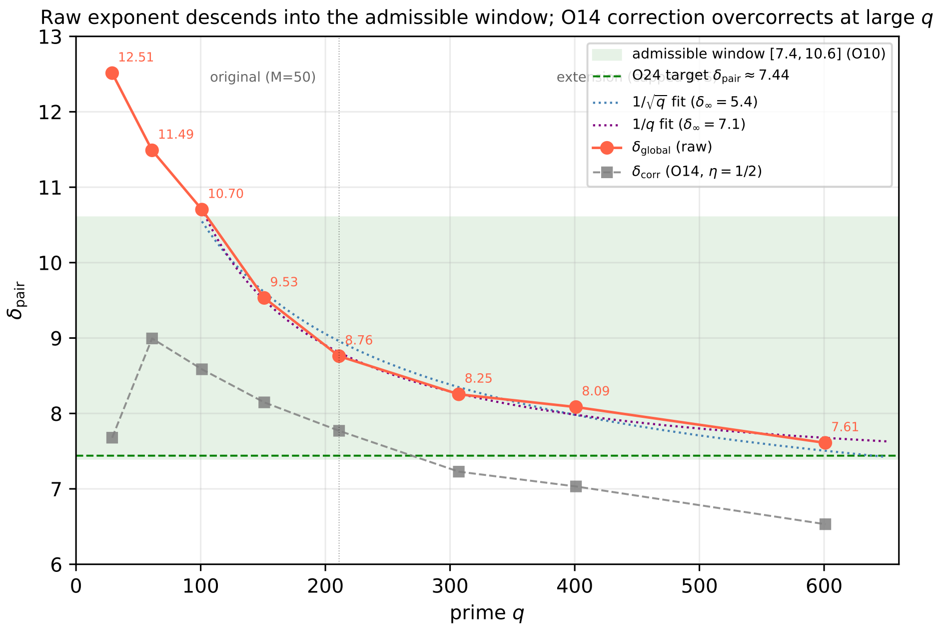

Figure 3: Extended campaign. Filled circles: raw $\delta_{\text{global}}(q)$, descending into the admissible window $[7.4, 10.6]$ (green band) and reaching 7.61 at $q = 601$. Grey squares: the O14-corrected δ_{corr}, which overshoots below the lower edge at large q. Dotted: $1/q$ and $1/\sqrt{q}$ fits on $q \geq 101$, both statistically viable, illustrating that δ_∞ is not pinned. The marker at $q = 211$ separates the original full-graph campaign from the capped-BFS extension.

4 Structural Analysis: Why Extrapolation Fails

4.1 Empirical fits are degenerate

The sequence $\bar{\delta}_{\text{pair}}(q)$ for $q \in \{61, 101, 151, 211\}$ is well described by multiple functional forms:

$$\bar{\delta}_{\text{pair}}(q) = \delta_\infty + \frac{a}{\log q}, \qquad \bar{\delta}_{\text{pair}}(q) = \delta_\infty + \frac{a}{(\log q)^2}, \qquad \bar{\delta}_{\text{pair}}(q) = \delta_\infty + \frac{a}{q^\alpha}. \quad (3)$$

All three fit the four data points with residuals below 0.07, while producing very different asymptotic values: $\delta_\infty \in \{4.89,\ 7.16,\ 5.45\}$ respectively.

This degeneracy is not a statistical artefact. It reflects a structural property of the problem: the leading correction behaves as $\log q / \log n_1(q)$, which remains close to unity over the accessible range since $n_1(q) = \Theta(q)$ (Window-Depth Theorem, O9 [8]). The deviations from δ_∞ are therefore indistinguishable from multiple slowly varying functions over the accessible range. The obstruction to extrapolation is not numerical precision, but the internal normalization geometry of the observable.

4.2 The leading correction behaves as $\log q / \log n_1(q) \approx 1$

The observable $\sigma_c(n)$ is normalised by $|S_n|$, whose growth in the pre-saturation window $[n_0, n_1]$ is not yet in the asymptotic Bass–Guivarc'h regime $|S_n| \sim n^3$. O14 [7] identifies the dominant finite-size correction as

$$\delta_{\mathrm{corr}}(q) = \delta_{\mathrm{pair}}(q) - \eta \, \frac{\log q}{\log n_1(q)}, \tag{4}$$

with $\eta = 1/2$ derived from the Weil amplitude scaling. Since $n_1(q) = \Theta(q)$, as established at the level of scaling behaviour in O9 [8] without requiring identification of the proportionality constant, one has $\log n_1(q) \sim \log q$ as $q \to \infty$, so the correction factor $\log q / \log n_1(q) \to 1$. From Table 1, the empirical values are 0.97–0.99 over the tested range, confirming that the correction is close to its limiting value. This identifies the BFS window depth as the effective scale controlling finite-size effects in the spectral observable. Any fit of the form $\delta_\infty + f(q)$ for a slowly varying f will describe the data well without carrying asymptotic meaning.

4.3 Corrected values are consistent with the admissible window

Applying (4) with $\eta = 1/2$ yields:

q	$\bar{\delta}_{\mathrm{pair}}$	$\eta \log q / \log n_1$	δ_{corr}
61	9.983	0.988	8.995
101	9.548	0.962	8.586
151	9.125	0.978	8.147
211	8.784	1.014	7.771

Table 4: O14 normalization correction applied to O25 data. The corrected values $\delta_{\mathrm{corr}}(q) \in [7.7, 9.0]$ fall within the admissible window $[7.4, 10.6]$ for all four primes. The correction factor $\eta \log q / \log n_1 \approx 0.96$–$1.01$ is approximately constant, as predicted by the $n_1 = \Theta(q)$ scaling.

The corrected values are consistent with an asymptotic value $\delta_\infty \in [7.4, 10.6]$, but do not uniquely determine it. We emphasise that $\eta = 1/2$ is imported from O14 and is not derived within the present pipeline; the correction should therefore be interpreted as a structural consistency test rather than a parameter-free prediction. The observation that it brings all four data points into the admissible window is consistent with this hypothesis but does not constitute a proof. Moreover, the large-prime extension of §3.4 shows that this correction is a *small-q* device: at $q \in \{307, 401, 601\}$ the raw exponent already lies in (or descends into) the admissible window, while the same correction overshoots *below* the lower edge (Table 3). The window

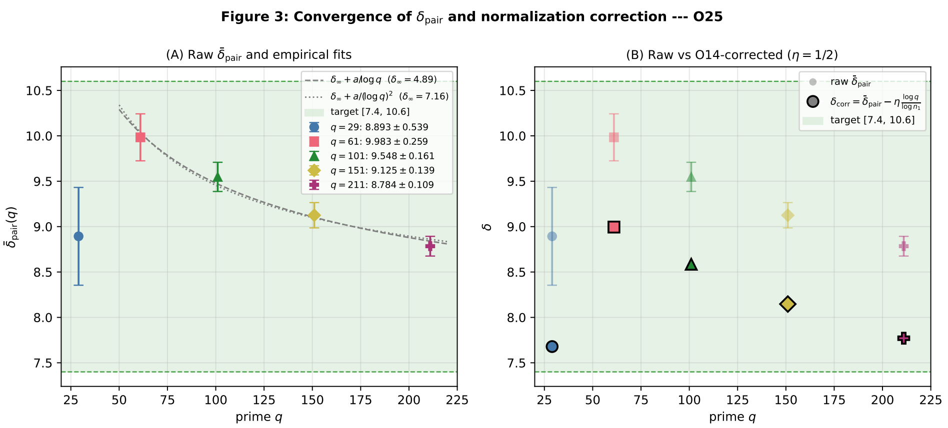

Figure 4: Left: raw $\bar{\delta}_{\rm pair}(q)$ with error bars $(\pm\sigma_q)$ and two empirical fits $(1/\log q$ and $1/(\log q)^2)$, both consistent with the data but producing different asymptotic values ($\delta_\infty \approx 4.89$ and 7.16 respectively), illustrating the structural degeneracy. Right: raw $\bar{\delta}_{\rm pair}$ (faded) vs O14-corrected $\delta_{\rm corr} = \bar{\delta}_{\rm pair} - \eta \log q / \log n_1$ (solid, $\eta = 1/2$). All corrected values fall within the admissible window $[7.4, 10.6]$ (green band).

agreement is therefore ultimately carried by the raw observable, and the O14 correction is best read as a finite-size adjustment that becomes unnecessary at large q.

5 The Central Open Direction: $n_1(q)/q$

The analysis of Section 4 identifies the ratio $n_1(q)/q$ as the key quantity controlling convergence. From Table 1: $n_1(q)/q \in \{0.138, 0.131, 0.109, 0.086, 0.066\}$ for $q \in \{29, 61, 101, 151, 211\}$. The ratio is *decreasing*, not yet constant. This means the system is not in the asymptotic regime $n_1 \sim \alpha q$ for any fixed constant α, and the correction of Section 4.2 is itself q-dependent.

Two sub-questions are open:

1. Does $n_1(q)/q$ stabilise as $q \to \infty$, and if so, to what value α?

2. If $n_1(q) = \alpha q + O(q^\beta)$ with $\beta < 1$, what is the resulting correction to $\bar{\delta}_{\rm pair}(q)$ at next-to-leading order?

Answering these questions would allow an analytical prediction of δ_∞ from the structure of $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, independently of numerical extrapolation. This also provides a structural explanation for the long-standing instability of $\delta_{\rm pair}$ estimates across primes observed since O12–O13 [3, 2]: the instability is not a measurement artefact, but reflects the unresolved q-dependence of $n_1(q)/q$.

6 Discussion

6.1 What O25 establishes

Three results are established:

1. **Numerical robustness.** $\delta_{\text{pair}}(q)$ converges and the inter-pair variance decreases strongly with q. At $q = 211$, the coefficient of variation is 1.2% over 105 pairs with $M = 50$ samples each. The decrease of σ_q is strictly monotone across all five primes with no sign of a plateau, strengthening the evidence that δ_{pair} is a structural invariant of the Weil representation. This confirms the structural prediction of O24 [15].

2. **Structural non-identifiability of δ_∞ by extrapolation.** The leading finite-size correction behaves as $\log q / \log n_1(q) \approx 1$, making all empirical fits degenerate over the accessible range. Direct extrapolation in q cannot identify the asymptotic value; the correct analytical variable is $n_1(q)/q$, not q itself.

3. **The raw observable reaches the admissible window; the O14 correction becomes unnecessary.** The extended campaign (q up to 601) shows that the raw exponent $\delta_{\text{global}}(q)$ descends into $[7.4, 10.6]$ on its own, reaching 7.61 at $q = 601$. The O14 normalization correction, which lifts the small-q values into the window, becomes progressively unnecessary and eventually overcorrects at large q (sending the corrected quantity below 7.4). The admissible-window agreement is therefore carried by the raw observable rather than by the corrected one — a stronger statement than the earlier small-q analysis, since it removes an ingredient rather than adding one.

6.2 Relation to O24

O24 [15] established the structural integrity of the chain $c_\chi \to \delta_{\text{pair}} \to \beta^*$ unconditionally. O25 provides the quantitative numerical content: the extended campaign shows that the raw δ_{pair} measured at large primes descends into the admissible window without correction, and that the inferred transfer parameter $\beta^* = 1/(\delta_{\text{pair}} + \frac{1}{2})$ stays in the narrow band 0.108–0.123 across $q \in \{211, \ldots, 601\}$ (Table 3), consistent with the O24 transfer analysis. This is a stability observation, not a new determination of β^*: since δ_∞ is not pinned, β^* inherits the same residual uncertainty. The two papers together close the numerical open direction of the O-series at the level of structural consistency.

6.3 What remains open

The following directions are explicitly open:

- Analytical derivation of the asymptotic ratio $n_1(q)/q$ as $q \to \infty$.
- Verification of the O14 hypothesis $\eta = 1/2$: the extension shows it overcorrects at large q, so the exact finite-size form (rather than the value $\eta = 1/2$) is the open question.
- Discrimination of the finite-size convergence form ($1/q$ vs $1/\sqrt{q}$), which the present range $q \leq 601$ does not resolve; this likely requires $q \gtrsim 10^3$.
- Execution of O26 Test 4 (effective dimension $r_{\text{eff}} = d_\rho^2$ of the covariance trajectory). A partial numerical implementation of the canonical embedding $\Phi_{q,\rho} = \rho \circ \iota \circ \pi$ (proved unconditionally in O27 [11]) was carried out on O25 data at $q \in \{29, 61\}$. The current approximation of π by the leading Gram–Schmidt basis vectors yields $r_{\text{eff}} = 2$ universally, an artefact of the DC mode dominating the projection; the correct implementation requires restricting the ADE eigenvectors of $\text{Cay}(G_q, S_q)$ to the Weil representation space via the Peter–Weyl coefficient map, identified as an open problem in Hecke spectral theory (see O26 [10], Remark 6.3).

7 Conclusion

The convergence of $\delta_{\text{pair}}(q)$ is robust and the inter-pair concentration is strong, confirming that δ_{pair} is a structural invariant of the admissibility programme. The central result of the extended campaign is that the *raw* exponent $\delta_{\text{global}}(q)$ descends monotonically into the admissible window $[7.4, 10.6]$ on its own, reaching 7.61 at $q = 601$, with no finite-size correction applied. The O14 normalization correction — required to bring the small-q values into the window — becomes progressively unnecessary and eventually overcorrects at large q, driving the corrected quantity below the lower edge; the admissible-window agreement is therefore carried by the raw observable, not by the corrected one. The asymptotic value δ_∞ remains insufficiently constrained by the accessible range: $1/q$ and $1/\sqrt{q}$ convergence laws are both statistically viable, so no single δ_∞ is claimed, and the discrimination of the finite-size form is left as the open numerical direction (likely requiring $q \gtrsim 10^3$). As a secondary consequence, the inferred transfer parameter $\beta^* = 1/(\delta_{\text{pair}} + \frac{1}{2})$ stays in the narrow interval 0.108–0.123 across $q \in \{211, \ldots, 601\}$, consistent with the O24 transfer analysis but inheriting the same residual uncertainty as δ_∞.

Data and code availability

The numerical pipeline `o25_paired_pipeline.py`, the checkpoint files `q{q}_o25.npz` for $q \in \{29, 61, 101, 151, 211\}$, the analysis script `o25_analysis.py`, and the scaling plot script `o25_scaling_plot.py` are available at 10.5281/zenodo.18292335. The extended M=20 run checkpoint files and the figure `o25_scaling.pdf` are included in the same deposit. The large-prime extension of §3.4 uses the capped-BFS campaign `delta_pair_convergence_campaign.py` (resumable, live ETA), the convergence analysis `delta_pair_analysis.py` (knee-trimmed interior fit, $1/q$-vs-$1/\sqrt{q}$ scan, bootstrap), and the checkpoints `dpc_q{307,401,601}.npz`, all in `simulation/spectral/o25`; the capped construction reuses `n1_resumable.build_bfs` and reproduces `compute_block_capacity` exactly.

Acknowledgements

The author thanks the extended discussions within the Cosmochrony programme that shaped the identification of $n_1(q)/q$ as the central asymptotic variable. AI-assisted tools were used during the preparation of this work. All scientific content, results, and conclusions are the sole responsibility of the author.

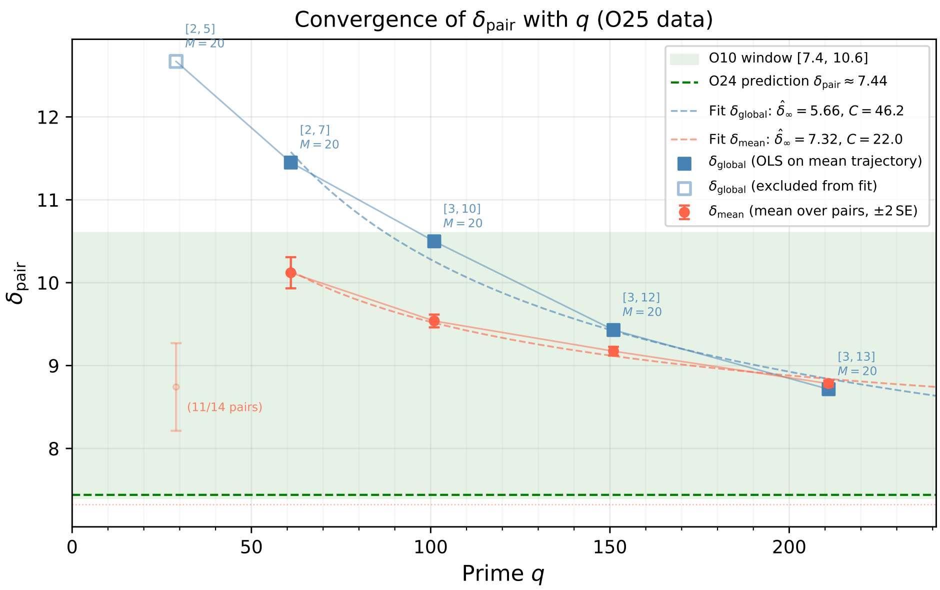

Figure 5: Convergence of δ_{pair} with q from the extended O25 campaign ($q \in \{29, 61, 101, 151, 211\}$, $M = 20$, windows from the WINDOW_O12 table). Filled symbols: primes included in the extrapolation fits ($q \in \{61, 101, 151, 211\}$). Open/faded symbols: $q = 29$, excluded from fits (only 11/14 pairs converge and the fitting window has 4 points; see §3.3 for the structural explanation). Squares (blue): δ_{global}, the OLS exponent on the mean σ_{pair} trajectory. Circles (red): $\bar{\delta}_{\text{pair}}$, mean over conjugate pairs, with $\pm 2\,\text{SE}$ error bars. Dashed red: fit $\bar{\delta}_{\text{pair}} \approx 7.32 + 22.0/\sqrt{q}$, extrapolating to $\hat{\delta}_\infty = 7.32$, within 0.12 of the O24 prediction of 7.44. Dashed blue: fit $\delta_{\text{global}} \approx 5.66 + 46.2/\sqrt{q}$, extrapolating to 5.66 (below O24, reflecting the known negative bias of δ_{global} relative to $\bar{\delta}_{\text{pair}}$). Green band: admissible window [7.4, 10.6] from O10. Green dashed: O24 prediction $\delta_{\text{pair}} \approx 7.44$. The single-form $1/\sqrt{q}$ extrapolation shown here (calibrated on $q \leq 211$) is *not* a determination of δ_∞: the large-prime extension of §3.4 (Figure 3) shows that $1/q$ and $1/\sqrt{q}$ laws remain equally viable, so no single asymptote is claimed.

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